



Application Lifecycle Management

Given the project risk and customer requirements, explain how to assess the benefits and risks of the different development methodologies and recommend the appropriate governance strategies based on the customer maturity.

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After studying this topic, you should be able to:

-  Describe the various stages of Application Lifecycle Management (ALM).
-  Explain the features, benefits, and risks of Waterfall, Agile, and Hybrid development methodologies.
-  Describe the principles of Scrum, Lean, and Kanban frameworks.
-  Apply governance strategies to development methodologies.



Introduction

Application Lifecycle Management (ALM) is the process of managing the life of an application. It manages an application's development, from design to final release. It establishes a framework for fixing bugs and delivering feature enhancements over time. It consists of these phases: **Plan Release, Develop, Test, Build Release, Test Release, and Release.**

Various **development methodologies** are available to manage the development of a Salesforce application. These include **waterfall, agile, and hybrid**. Furthermore, the principles of frameworks such as **Scrum, Lean, or Kanban** can also be applied to a Salesforce development project.

It is also important to consider the establishment of **governance strategies** for a development methodology to manage and track changes effectively.



Application Lifecycle Management & Development Methodologies

Application Lifecycle Management

Application Lifecycle Management (ALM) is the process of managing the life of an application.

Waterfall

Waterfall is a traditional development methodology with a linear sequence of defined steps.



Agile

Agile is an iterative and flexible development approach that focuses on incremental delivery.



Scrum

The Scrum framework focuses on the iterative delivery of results based on five core values.



Lean

The Lean framework applies the principles of efficiency and value delivery to software development.



Kanban

Kanban is an agile framework that focuses on managing and improving workflows.



Governance

Governance refers to putting a structure in place to handle and track changes effectively.



Governance Strategies

Governance strategies can be implemented for methodologies such as waterfall and agile.



Application Lifecycle Management

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Application Lifecycle Management

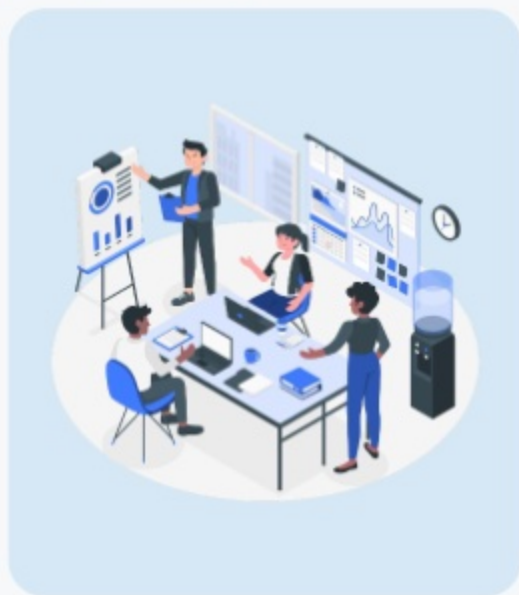
Application Lifecycle Management (ALM) refers to the **process of managing the life of an application** and focuses on the **steps** and **policies** to build the application smoothly.

❖ STAGES

The **Application Lifecycle Management (ALM)** cycle consists of the following stages that apply to any Salesforce development project: **Plan Release, Develop, Test, Build Release, Test Release, and Release**

❖ PROCESS & FRAMEWORK

ALM defines the process of managing an app's development, from **design to final release**. It also establishes a framework for **fixing bugs** and **developing feature enhancements** over time.



Application Lifecycle Management

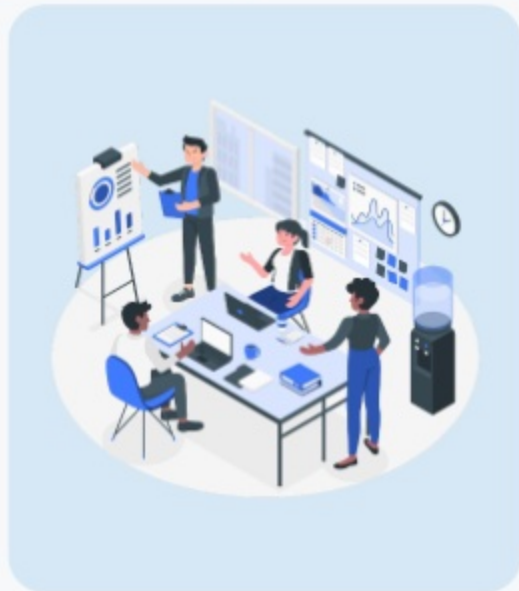
Application Lifecycle Management (ALM) refers to the **process of managing the life of an application** and focuses on the **steps** and **policies** to build the application smoothly.

❖ PLAN RELEASE

A development project should start with a **plan**. At this stage, **requirements should be gathered** and **analyzed**. The product manager creates **design specifications**. The **development** and **testing environments** are identified.

❖ DEVELOP

At this stage, based on the design specifications, the **development work is done** in an environment containing a copy of the production org's metadata. An appropriate combination of **declarative and programmatic tools** is used for development.



Application Lifecycle Management

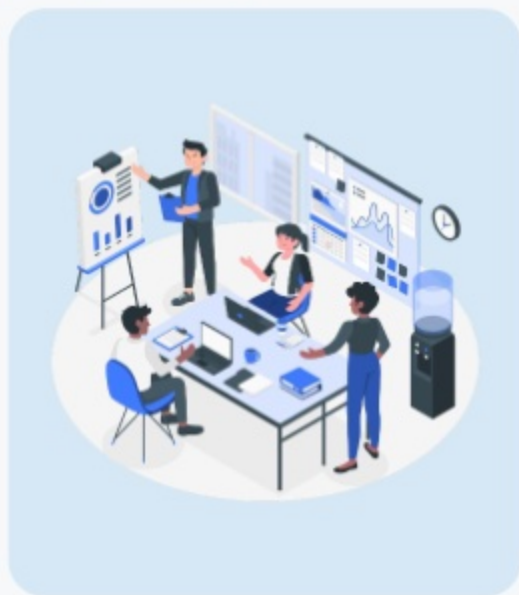
Application Lifecycle Management (ALM) refers to the **process of managing the life of an application** and focuses on the **steps** and **policies** to build the application smoothly.

❖ TEST

The changes made in the previous step are **tested** to ensure they work as intended before they are **integrated with other people's work**. Testing is done in the same type of environment that was used for development.

❖ BUILD RELEASE

The customizations that need to be deployed to production are **aggregated into a single-release artifact**.



Application Lifecycle Management

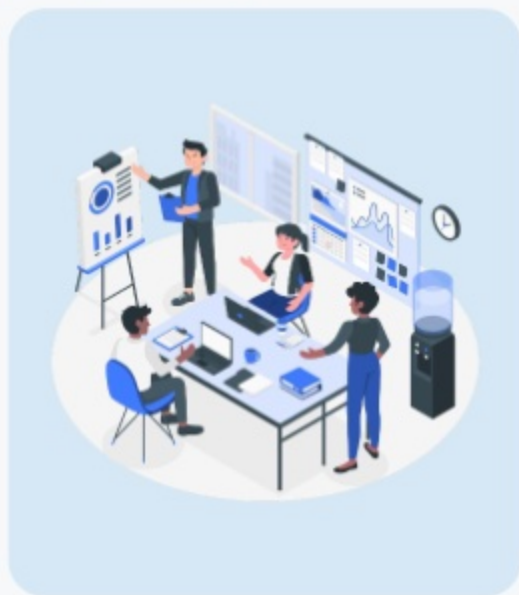
Application Lifecycle Management (ALM) refers to the **process of managing the life of an application** and focuses on the **steps** and **policies** to build the application smoothly.

❖ TEST RELEASE

The **release artifact is tested in a staging environment** that mimics production. **Regression and performance tests** are run at this stage. The release is tested with a small set of experienced people who provide feedback, called **user acceptance testing (UAT)**.

❖ RELEASE

After **completing the testing** and **meeting the quality benchmarks**, the release artifact containing the customizations is **deployed to production**. **Employees and partners are trained** to ensure that they understand the changes, and if necessary, a separate environment is created for training if the release has a significant user impact.



Application Lifecycle Management



Benefits of Application Lifecycle Management

Using **Application Lifecycle Management (ALM)** for application development has many benefits.

◆ AVOIDS DEFECTS

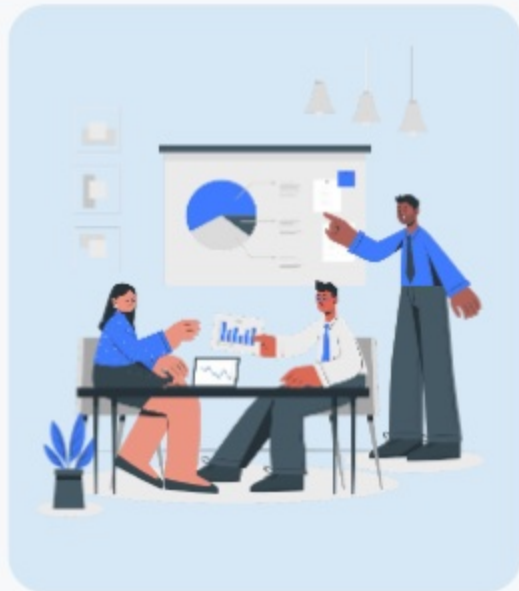
Using ALM **improves the quality** of the application and ensures that **defects**, such as functional bugs, usability issues, and logic errors, **are not deployed to production**.

◆ INCREASES EFFICIENCY

It **increases the efficiency and productivity of teams** by standardizing processes and facilitating the reuse of resources.

◆ IMPROVES CONSISTENCY

It enables systematic planning, development, and deployment, leading to **more consistent and predictable release schedules**.



Benefits of Application Lifecycle Management

Using **Application Lifecycle Management (ALM)** for application development has many benefits.

◆ FASTER DELIVERY

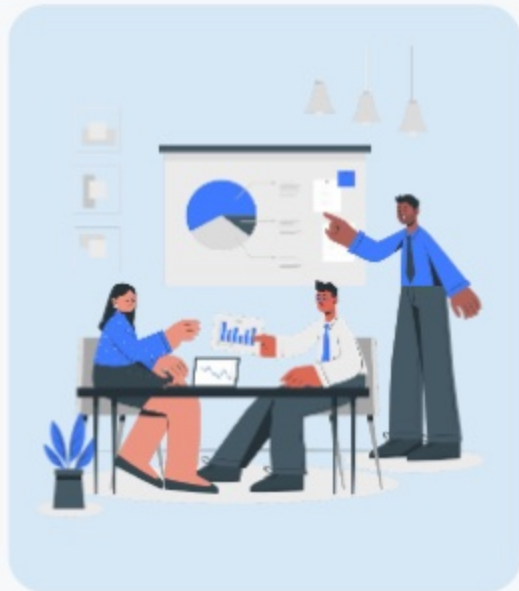
It can **accelerate the development process**, enabling teams to deliver features to customers faster.

◆ REDUCES COST

It can **significantly reduce costs** by improving efficiency, reducing rework, and preventing defects.

◆ MAINTENANCE

It provides a framework for **ongoing maintenance and support of the application**, ensuring **long-term stability and usability**.



Development Methodologies

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Development Methodology

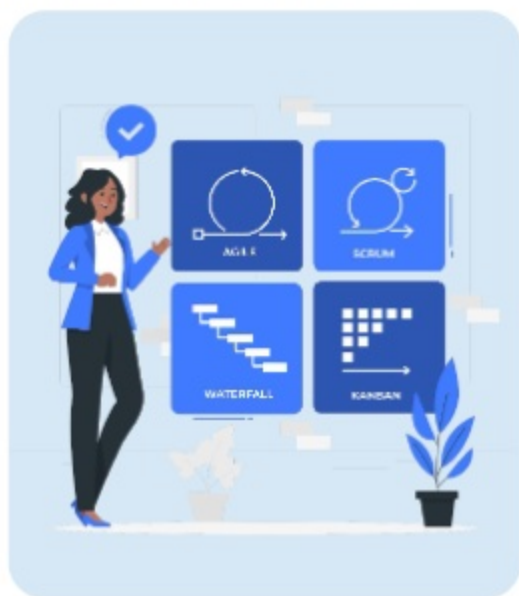
A **development methodology** is the **structured approach or set of practices** that Salesforce developers and teams follow to **build, test, deploy, and maintain** applications within the Salesforce platform.

◆ PROCESSES & STRATEGIES

The development methodology encompasses the **processes and strategies** used to manage the lifecycle of custom Salesforce solutions, through **initial planning** and **design** through **development, deployment,** and **ongoing maintenance**.

◆ KEY ASPECTS

The key aspects of a development methodology are **planning and analysis, design, development, testing, deployment, maintenance and support,** and **governance**.



Waterfall Methodology

Waterfall is a **traditional development methodology** with a **linear sequence of defined steps**. Each step is completed fully before moving on to the next.



The **requirements** are **well-defined** and typically **very stable**.



There is a focus on **extensive planning and documentation**.



There is a **single outcome or result** of the development process.



Stakeholders **engage in planning** and at **key milestones**.

Waterfall Methodology

Waterfall is a **traditional development methodology** with a **linear sequence of defined steps**. Each step is completed fully before moving on to the next.



Time and cost are determined in the **planning** stage of the project.



The **project risks** are defined **early** in the project.



Changes are carefully **managed and controlled**.

Risks of Waterfall Methodology

The **Waterfall development methodology** has various **risks**.

There is a **lack of flexibility** and the changes must be **formally reviewed and approved**.

Change is avoided and new or unplanned **risks are not easily accommodated**.

The documentation is **time-consuming** and **doesn't provide value** to the customer.

Stakeholder involvement is **minimal**, so there is **limited feedback**.

Accurate and complete planning is required to determine the project time and cost.

There is a **risk of customer dissatisfaction** with the final outcome.

Agile Methodology

Agile is a **less traditional** development methodology that focuses on an **iterative and flexible approach** where the project is divided into small parts called **sprints**.



The **requirements** are progressively elaborated.



The focus is on **iterative planning and execution**.



There is **incremental delivery of outcome**.



Stakeholders **engage throughout the project**.

Agile Methodology

Agile is a **less traditional** development methodology that focuses on an **iterative and flexible approach** where the project is divided into small parts called **sprints**.



As the requirements are elaborated, the **time** and **cost** are **defined** and **refined**.



Project risks are **defined** and **refined throughout the project**.



Changes are **expected** and **incorporated** during **planning and execution**.

Risks of Agile Methodology

The **Agile development methodology** has various **risks**.

Projects can suffer from **scope creep** due to the iterative nature and frequent feedback loops.

Rapid development can sometimes **compromise the quality** of software.

Prioritizing working software can lead to **insufficient documentation**.

Excessive customer input can **derail the project focus** or lead to **constant changes** in requirements.

Lack of detailed upfront planning can cause **underestimation of time and effort**.

Integration can be challenging if multiple teams are working on different parts of a larger system.

Agile Methodologies

The three most common Agile methodologies are **Scrum**, **Lean**, and **Kanban**.

AGILE METHODOLOGIES		
 SCRUM	 LEAN	 KANBAN



Scrum

Scrum is an **agile framework** used for the **iterative delivery** of **project outcomes and results**. It is an **easy to follow** and **scalable** method often used for **software development**.

INCREMENTAL

Working software is **delivered incrementally**.
Time-boxed iterations known as **sprints**
typically last for 2-4 weeks.



MEETINGS

Scrum features meetings such as **sprint planning**, **daily stand-ups**, **sprint review**, and **sprint retrospective**.



ROLES

Stakeholders and **team** are **highly collaborative**. Roles include **scrum master**, **product owner**, and **development team**.



ARTIFACTS

Major artifacts include **product backlog**, **sprint backlog**, **burndown charts**, and **increment**.



Scrum Values

Openness

The team and stakeholders are transparent about all the work and any challenges. They share ideas, concerns, and mistakes. They are open to change, feedback, and learning.

Commitment

Team members are committed to the success of the project and the goals of the sprint. They are dedicated to the necessary time and effort to complete tasks.

Respect

Team members recognize each member's expertise, perspective, and contribution. They treat each other with dignity, work together as equals, and value their input.

Focus

Team members concentrate on the overall goals and work of the sprint. They prioritize the work that contributes to the sprint goals and avoid distractions.

Courage

Team members have the bravery to do the right thing. They are bold in the face of challenges and do not hesitate to speak up, experiment, or innovate.



Lean

Lean is a **framework** that applies the **manufacturing principles of efficiency and value delivery** to software development.

WASTE ELIMINATION

Lean focuses on **maximizing value** while **minimizing waste** associated with unproductive tasks.



KNOWLEDGE

Constant learning and adaptation are encouraged for continuous improvement.



EFFICIENCY & QUALITY

There is an **efficient** use of **team resources**. The focus is on **quality** at **every step**.



JUST-IN-TIME DECISIONS

Decisions are made at the **last responsible moment**. **Up-front decisions** are avoided.



Lean Principles

Identify value

The focus is on determining what is valuable to the customer by understanding their needs and preferences through engagement.

Map the value stream

Value stream mapping focuses on identifying the activities and processes necessary to deliver a feature and eliminating waste.

Create flow

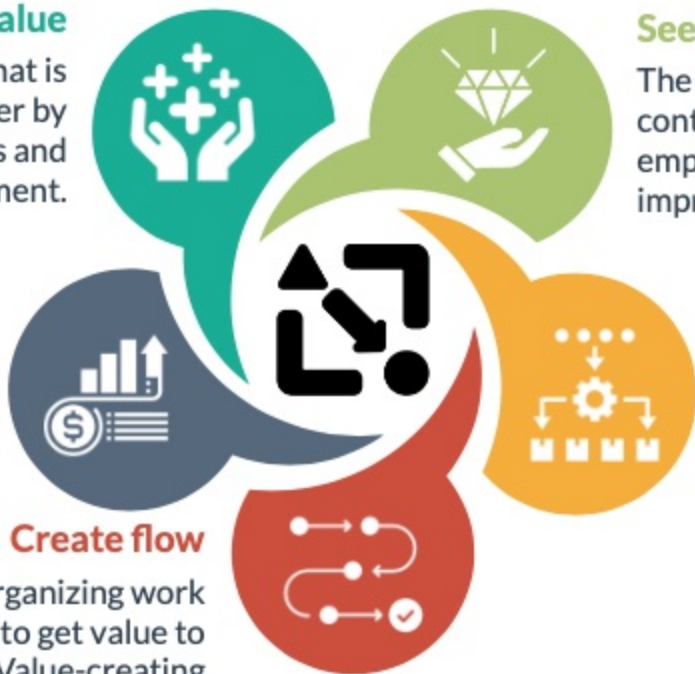
Creating a flow involves reorganizing work processes, equipment, and teams to get value to flow to customers at a steady pace. Value-creating steps are performed in a tight sequence.

Seek perfection

The final characteristic of Lean is the continuous pursuit of perfection. It emphasizes continually seeking ways to improve processes and eliminate waste.

Establish pull

Features are only made in response to customer demand. This means not delivering features unless there is a clear need or request.



Kanban

Kanban is an **agile framework** that focuses on **managing and improving workflows**.

VISUAL WORKFLOW

Work is made **visible**, typically using a **Kanban board**. The workflow is divided into **columns**, each representing a **different stage**.

CHANGES

Kanban allows teams to embrace **smaller and more changes** along the workflow.

LIMITS

Limitations are placed on how many **work items are in progress** at a time in **each workflow stage**.

METRICS

Metrics such as **cycle time** and **throughput** are used to determine and predict progress.



Kanban Traits

Visualize workflow

Work is divided into smaller pieces that are represented on cards that move from left to right on a Kanban board, which visually maps out each stage of the workflow.

Incremental & evolutionary change

Teams using the Kanban workflow are more open to taking last-minute requests. When a work item is finished, they move to the next highest-priority task.

Limit work in progress (WIP)

WIP limits are set at each stage of the workflow to prevent overloading the team and to ensure the smooth flow of work through the process. Current tasks are completed before starting new ones.

Kanban metrics

Cycle time is used to determine the number of days to complete a work item once work starts on it. Throughput is used to measure the amount of work completed in a single period of time.



Hybrid Methodology

A **hybrid development methodology** combines the **elements of both Waterfall and Agile methodologies** by leveraging their strengths and mitigating their weaknesses.



Structured phases, such as detailed upfront planning, are combined with the **flexibility** of Agile, such as iterative development.



Initial planning and **requirement gathering** are done in a **detailed and structured manner**.



Planning is followed by the use of **iterative development cycles** for implementation.



Regular feedback is incorporated while maintaining **defined milestones** and **deliverables**.

Hybrid Methodology

A **hybrid development methodology** combines the **elements of both Waterfall and Agile methodologies** by leveraging their strengths and mitigating their weaknesses.



Comprehensive **documentation** is **balanced** with minimal but essential documentation to ensure clarity without overburdening.



Risk assessment in early phases is combined with the ability to **adapt to risks as they arise**.



Waterfall techniques are often used for the **predictive aspects**. Agile techniques are applied to **uncertain parts**.

Risks of Hybrid Methodology

The **Hybrid development methodology** has various **risks**.

Balancing the practices of the two methodologies can be **complex** and **lead to confusion**.

The different nature of the two methodologies can lead to **conflicts**.

Implementing both the methodologies can **increase the administrative overhead** and **strain resources**.

If the guidelines are unclear, teams may **struggle to determine when to apply Agile principles** and Waterfall practices.

Teams **accustomed** to one methodology may find it **challenging** to adopt a combined approach.

Combining the two methodologies can lead to **suboptimal outcomes** due to **potentially diluted benefits**.

Governance Strategies

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Governance

Governance refers to putting a **structure** in place to effectively **handle and track changes** to all the aspects that control how applications work.

❖ APPLICATION

Governance can apply to the **development of custom applications** by the business in Salesforce and the **introduction of release updates** three times a year by Salesforce.

❖ NEED

To adapt to the changing business climate, organizations need **consistent tweaks** to **how changes are managed**. It can be a new system or process to implement, or a new dataset to track.



Governance

Governance refers to putting a **structure** in place to effectively **handle and track changes** to all the aspects that control how applications work.

❖ FOCUS

Governance frameworks generally focus on aspects such as **release management**, **prioritization of change requests**, **approval of changes**, **release schedule**, **communication process**, etc.

❖ BENEFITS

Benefits of using governance strategies include **increased efficiency** due to better resource utilization, **improved security and compliance**, **greater collaboration and transparency**, etc.



Waterfall Development Governance Strategies

Governance strategies for the **Waterfall development methodology** include the establishment of a **Change Control Board (CCB)** and setting up **Milestone Reviews**.



◆ CHANGE CONTROL BOARD

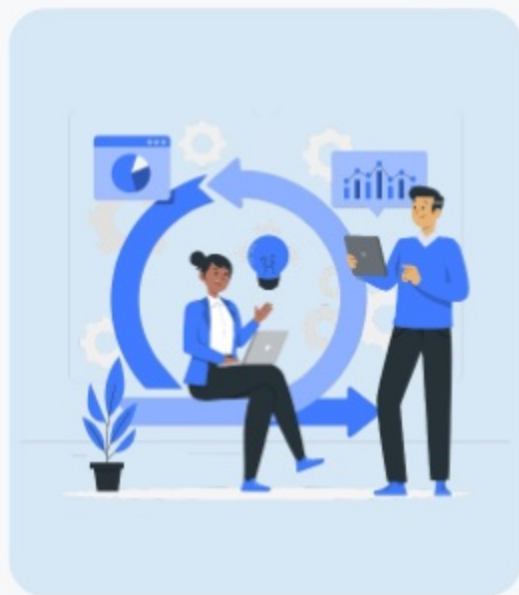
A **Change Control Board (CCB)** formal group that **assesses and approves changes** to the **project scope**, ensuring that they align with the original goals and objectives.

◆ MILESTONE REVIEWS

Meetings can be scheduled regularly to **review progress** and ensure that the **project stays on track** with its **timeline and budget**.

Agile Development Governance Strategies

When using the **Agile development methodology**, some governance practices include **prioritization**, **daily stand-ups**, **sprint reviews**, and **retrospectives**.



❖ WORK PRIORITIZATION

The **product owner** prioritizes **work items** and ensures alignment with customer needs and organizational goals.

❖ DAILY STAND-UPS

Teams can use **short, daily meetings** to **report progress** and **raise issues**.

❖ REVIEWS & RETROSPECTIVES

Regular meetings are conducted at the end of each sprint to **review work** and **discuss ways to improve processes**.

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Software Inc. is a global consulting company with offices worldwide that provides Salesforce consulting services to other companies. Its offerings include product implementation, application development, customization, and training delivery.

The company has several development teams, one for each region. For example, the AMER team handles implementation and development services for businesses in North America.

The development teams often handle unique project situations with certain challenges or risks. A Salesforce Architect has been hired to help the teams adopt the most suitable development methodologies and recommend related governance strategies after assessing the benefits and risks.

Requirements & Solutions

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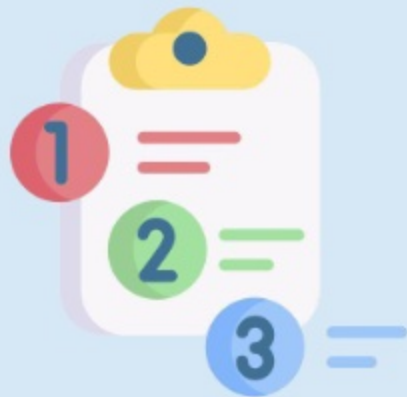


Requirement 1 (Part 1)



SCENARIO

The AMER development team of the company is tasked with developing a custom Salesforce application for a large enterprise client in the United States. The client has a **well-defined project scope** with **clear, stable requirements** and a **fixed deadline**. The project involves integrating Salesforce with several legacy systems. The client, being in a heavily regulated industry, requires **comprehensive documentation** for compliance for every phase of the project. The client's internal team is accustomed to traditional project management, prefers a **structured methodology**, and has **limited availability** for frequent engagements.

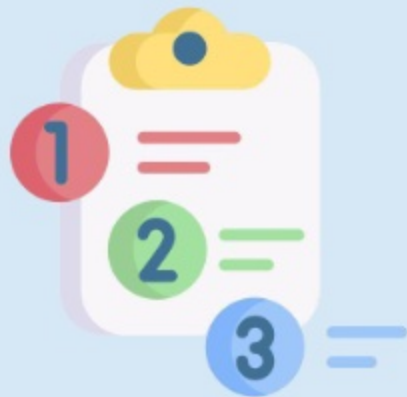


Requirement 1 (Part 2)



SCENARIO

The Salesforce Architect at Cosmic Software Inc. needs to recommend a suitable development methodology for this project, considering the benefits and risks, while taking into account the client's maturity in terms of project management and compliance requirements.



Solution 1 (Part 1)



SOLUTION

The **Waterfall** development methodology should be adopted for this project for several reasons. It uses a **linear and sequential approach** that provides a **clear structure** and a **defined pathway** from project initiation to completion, which aligns with the client's preference for a structured methodology. The project's well-defined requirements and scope are ideal for this methodology. The **extensive documentation** required in Waterfall can meet the client's compliance needs. The fixed deadline and limited availability for engagement also align well with the Waterfall approach of **set timelines and deliverables**.



Solution 1 (Part 2)



SOLUTION

However, some risks are associated with using the Waterfall methodology, but they can be mitigated using some strategies. It can be challenging to incorporate changes late in the project, which can be avoided through **exhaustive requirement gathering** and stakeholder sign-off before proceeding to development. A **feasibility study** and **architectural evaluation** should be conducted early in the project to mitigate late testing and integration risks. In addition, **milestone review meetings** should be conducted to ensure that there are no gaps in expectations.



Solution 1 (Part 3)

Waterfall Development Methodology



Requirement 2 (Part 1)



SCENARIO

A rapidly growing e-commerce startup in the United Kingdom has approached the company for a Salesforce implementation project. The startup is looking to implement Salesforce to manage its sales, customer service, and marketing activities. It currently uses a standalone legacy system to manage CRM processes and data. The chief executives value **flexibility**, as they anticipate their needs evolving quickly due to their **fast-paced and dynamic market**. They've also expressed a desire for frequent updates and the ability to **adjust features** based on **real-time user feedback**.



Requirement 2 (Part 2)



SCENARIO

Given the client's needs, the Salesforce Architect at Cosmic Software Inc. needs to evaluate the suitability of the Agile methodology for this project. This evaluation must consider the benefits and risks of the methodology in the context of the client's rapidly changing requirements and their relative maturity with agile project management practices.



Solution 2 (Part 1)



SOLUTION

The **Agile** development methodology is suitable for the evolving needs of the startup, as it allows for **frequent reassessment** and **adaptation of requirements**. The **iterative** nature of Agile enables the delivery of work in **small, manageable increments**, allowing the startup to begin using functionalities sooner and provide ongoing feedback. Moreover, Agile emphasizes **collaboration** between the development team and the client, ensuring that the final product closely aligns with the client's expectations and needs.



Solution 2 (Part 2)



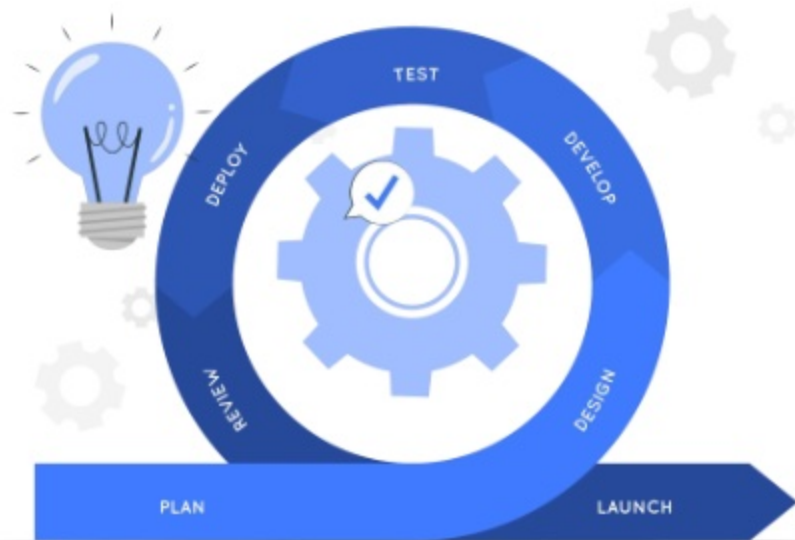
SOLUTION

However, certain risks and related mitigation strategies need to be considered for using the Agile methodology. If not managed carefully, the flexibility offered by Agile can lead to **scope creep**. A **robust backlog prioritization** process should be implemented, and changes in scope should be systematically evaluated and agreed upon. Since Agile requires active client involvement, it must be ensured that the client is committed to this level of participation. **Regularly scheduled meetings** and **demos** can facilitate this. Rapid delivery can compromise the depth of initial planning, which can be mitigated by **sprint planning sessions** that clearly outline goals and deliverables for each iteration.



Solution 2 (Part 3)

Agile Development Methodology



Requirement 3 (Part 1)



SCENARIO

A mid-sized retail company in Australia has been using Salesforce for basic functions but now wants to expand and customize the platform to support more complex operations, including integration with other enterprise systems. The company's management is **familiar with traditional project management methodologies** but is also open to **more modern, iterative approaches**. They seek a **balance** between a **structured plan** (due to the complex nature of integration) and the **flexibility** to adapt to user feedback and changing market trends.



Requirement 3 (Part 2)



SCENARIO

The Salesforce Architect at Cosmic Software Inc. needs to recommend a suitable development methodology for this project. The decision must consider the benefits and risks of the methodology in the context of the company's moderate maturity level in project management and their specific requirements for the Salesforce customization project.



Solution 3 (Part 1)



SOLUTION

The **Hybrid** development methodology should be adopted for the project since it provides the **structured planning and clear milestones of Waterfall**, which are essential for complex system integration, while also incorporating the **flexibility and adaptiveness of Agile** for gathering and implementing user feedback. The combination of early risk identification and mitigation and ongoing risk assessment would be beneficial for integrating Salesforce with multiple enterprise systems. Agile practices can be gradually introduced, ensuring stakeholder comfort and buy-in. Agile sprints can be employed for the development of custom features and for gathering user feedback.



Solution 3 (Part 2)



SOLUTION

On the other hand, mixing the two methodologies can lead to confusion and conflict, which can be avoided by **providing training** to the team and **clearly defining** which parts of the project will use Waterfall and which will use Agile. There's a risk that this methodology can use more resources, so **careful planning and resource management** are necessary to ensure that the project stays within the budget and time constraints. **Regular meetings and clear communication** should be used for a smooth transition between Waterfall and Agile phases. Additionally, **phase-gate reviews** should be implemented at major milestones for quality and progress checks.



Requirement 4 (Part 1)



SCENARIO

A healthcare services provider in Japan needs a custom Salesforce application to streamline patient management and internal operations. The provider is **new to Agile methodologies** but has expressed interest in adopting an Agile approach. Their organizational culture is deeply rooted in values such as **commitment** to patient care, **patient-first approach**, **openness** and **transparency** in operations, and **respect** for both patients and staff. They are looking for a development process for its team that is **flexible** and can **adapt** to their dynamic healthcare environment while maintaining these core values.



Requirement 4 (Part 2)



SCENARIO

The Salesforce Architect needs to assess and recommend whether the Scrum framework is suitable for this Salesforce application development project. The assessment must consider the alignment of **Scrum's five values (Commitment, Courage, Focus, Openness, and Respect)** with the healthcare provider's organizational culture. Additionally, the assessment should identify potential risks associated with adopting the Scrum methodology in this context and propose strategies to mitigate these risks. The goal is to ensure that the chosen development methodology not only aligns with the provider's values but also effectively supports their transition to Agile practices.

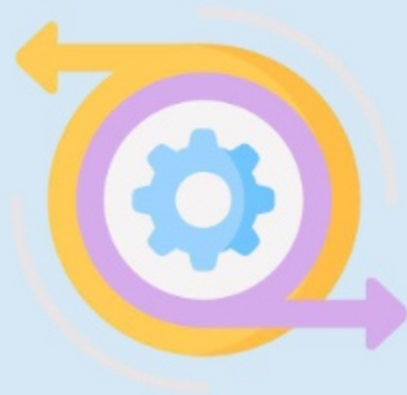


Solution 4 (Part 1)



SOLUTION

The healthcare services provider's dedication to improving patient care indicates a strong alignment with the Scrum value of **commitment**. The provider's openness to Agile methodologies demonstrates **courage** to innovate and adapt. The focused approach to patient services (patient-first) aligns with Scrum's emphasis on maintaining **focus** on the sprint and project objectives. The openness and transparency in operations aligns well with the Scrum value of **openness** in processes and communication. Lastly, the respect for the provider's patients and staff aligns with the Scrum value of **respect** for team members and their contributions.

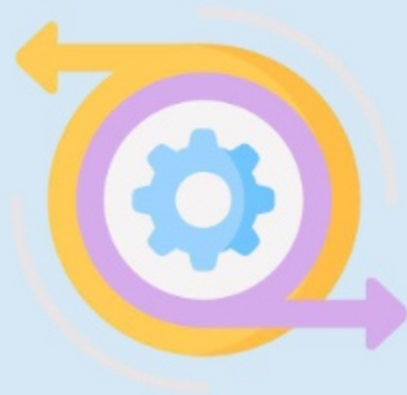


Solution 4 (Part 2)



SOLUTION

Although the provider's organization culture aligns with the five values of Scrum, there are certain risks associated with each. There is a potential risk in translating commitment to the iterative development process, which may be new to the provider. Ensuring **consistent engagement** and **dedication to Scrum ceremonies and Sprint goals** would be crucial. **Continuous support and resources** should be provided to help the team navigate the Scrum framework. There can be reluctance or resistance in the face of complex or unfamiliar Agile practices. Managing this change and **fostering a culture that embraces Agile challenges** would be important.

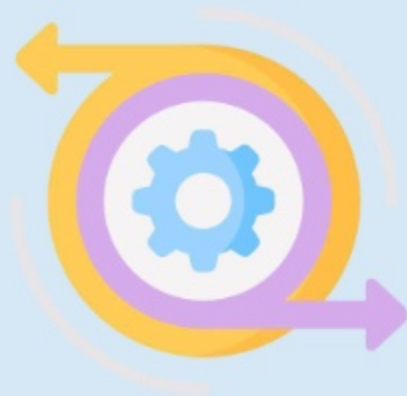


Solution 4 (Part 3)



SOLUTION

There can be potential distractions or loss of focus due to the day-to-day demands of healthcare operations. It would be necessary to ensure that the Scrum team is able to **focus on sprint tasks without any interruptions**. As far as openness is concerned, the team may find it challenging to adapt to the high level of transparency required in Scrum, especially in sharing progress, challenges, and feedback openly. This risk can be mitigated by **providing training on constructive feedback**. Moreover, there may be differing opinions on how tasks should be approached, leading to potential conflicts, which can be avoided through **team-building activities**.



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-  [Technical Project Management](#)
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